

CV 5101
Modeling, Uncertainty, and
Data for Engineers
(July – Nov 2026)

Dr. Prakash S Badal

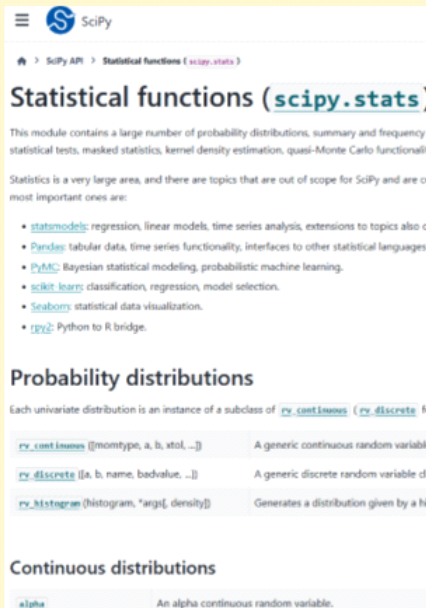
Flow

- Covariance
- Correlation
- Parametric distributions
- SciPy

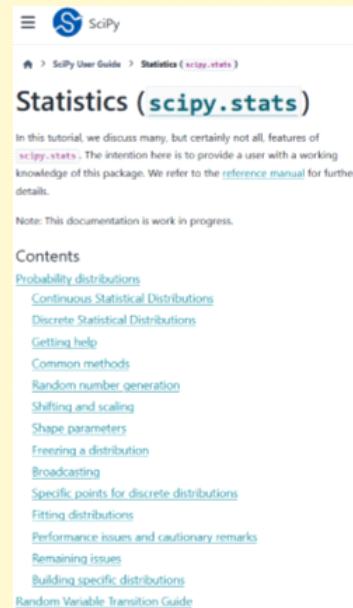
L5
Sep 9, 2026

Announcement

- Practice `scipy.stats` for the following:
 - Using distributions → Shifting, scaling
 - Generating random numbers
 - Fitting distributions



The screenshot shows the SciPy documentation page for Statistical functions (`scipy.stats`). The page title is "Statistical functions (`scipy.stats`)". Below the title, there is a brief description of the module and a list of submodules: `statmodels`, `Pandas`, `PyMC`, `scikit-learn`, `Seaborn`, and `ipy`. The page also includes a section for Probability distributions, which lists `rv_continuous`, `rv_discrete`, and `rv_histogram`. The page is part of the SciPy API documentation.



The screenshot shows the SciPy documentation page for Statistics (`scipy.stats`). The page title is "Statistics (`scipy.stats`)". Below the title, there is a brief description of the module and a list of submodules: `statmodels`, `Pandas`, `PyMC`, `scikit-learn`, `Seaborn`, and `ipy`. The page also includes a section for Probability distributions, which lists `rv_continuous`, `rv_discrete`, and `rv_histogram`. The page is part of the SciPy User Guide documentation.

scipy

- [Ref.] <https://docs.scipy.org/doc/scipy/reference/stats.html>
- Tutorial <https://docs.scipy.org/doc/scipy/tutorial/stats.html>
- Probability distributions
https://docs.scipy.org/doc/scipy/tutorial/stats/probability_distributions.html
- Run examples

- Common Methods
- Random number generator
- Shifting-scaling: loc, scale
- Fitting distributions

Check-in with teach-book

<https://mude.citg.tudelft.nl/book/2024/>

- Q1 Topics (Chapters 5, 6, 2, and 3)
 - Q1C5 Univariate continuous distribution
 - Q1C5.1 PDF/CDF
 - Q1C5.2 Empirical Distributions
 - Q1C5.3 Parametric Distributions
 - Q1C5.4 Fitting a Distribution
 - Q1C6 Multivariate Distributions (briefly)
 - Q1C2 Propagation of Uncertainty
 - Q1C3 Observation Theory: least-sq., Hyp. Test, Conf. Intervals
- Q2 Topics (Chapter 7 and 8)
 - Q2C7 Extreme value theory: GEV, return period, POT
 - Q2C8 Risk and decision making (CBA)
- Fundamental Concepts
 - Chapter 6, 7, 8, 9. Probability basics, rv, z- and t-tables
- Programming
 - Fundamental Concepts → Chapter 10

6

Module Overview

B.1

Uncertainty & Estimation

B.1.1

Probabilistic basics and RVs

- Probability laws
- Random variables
- Discrete, continuous RVs
- PMF/PDF/CDF
- Expectation
- Variance/cov./corr.

Q1C5, Funda6-7-8

B.1.2

Core distributions

- Gaussian
- Uniform
- Exponential
- Lognormal
- Symmetric/skewness
- *Tails*: thick/light

Q1C5, Funda9-10

B.1.3

Propagation laws & least-sq.

- Uncertainty propagation
- Least-square errors
- Parameter estimation

Q1C2-C3, Funda9-10

B.1.2 Core distributions (Q1C5)

Normal distribution: motivation

Human Height

- Genetics
- Nutrition during 1000s of childhood
- Hormone production
- Sleep quality
- Skeleton response to mechanical stress

- Various additive effects
- Ubiquitous
- Mathematically tractable

Concrete strength

- Water-cement ratio
- Exact grain-size distribution
- Impurities
- Micro air pockets

SAT score:

-

Normal (Gaussian) PDF

$$f_X(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



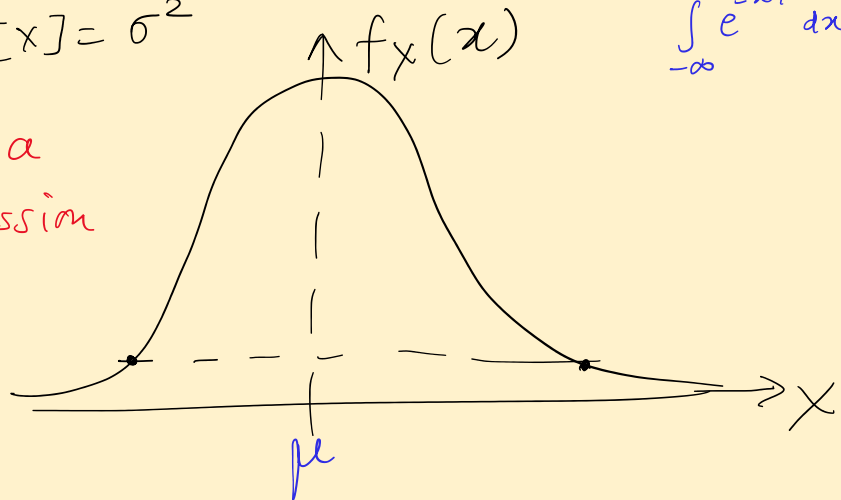
Gaussian/
Normal/
Bell-shaped

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

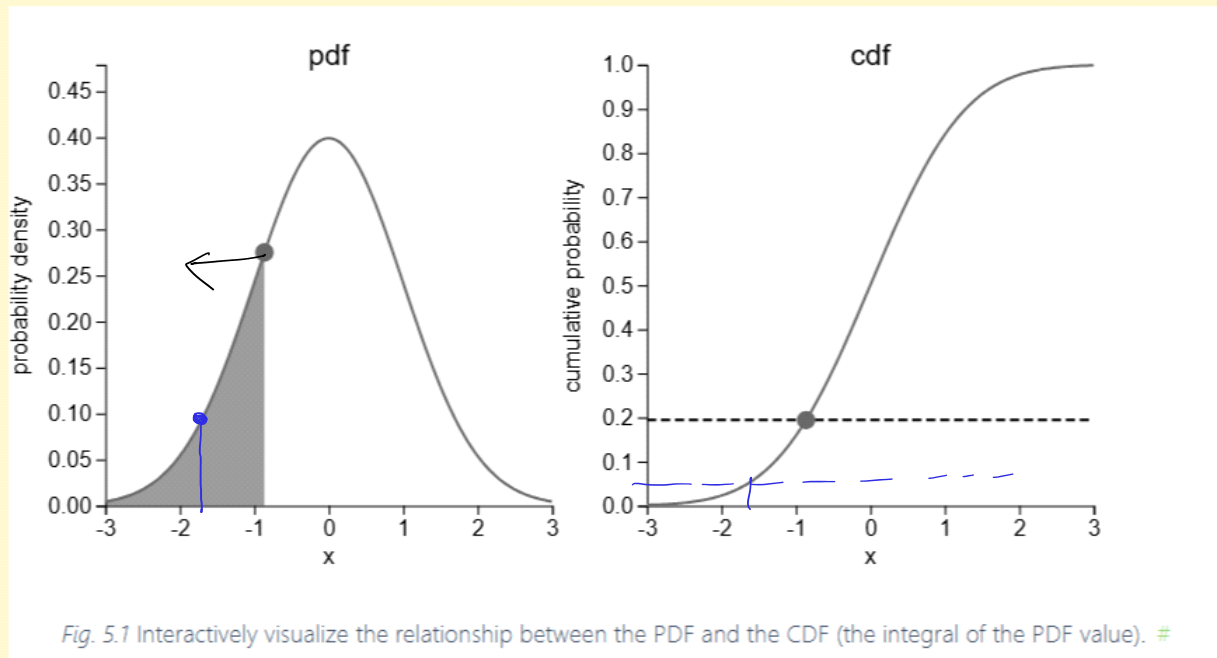
$$E[X] = \mu \quad ; \quad \text{Var}[X] = \sigma^2$$

CDF does NOT have a
closed-form expression

$$\int_{-\infty}^{\infty} e^{-x^2/2} dx = \sqrt{2\pi}$$



Gaussian



TeachBook Fig-5.1
Q1-CS

Standard Normal Probabilities

Z-table

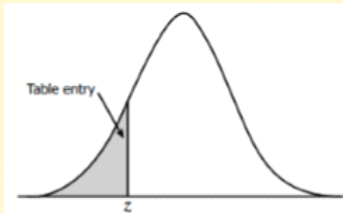
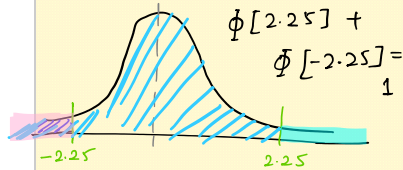
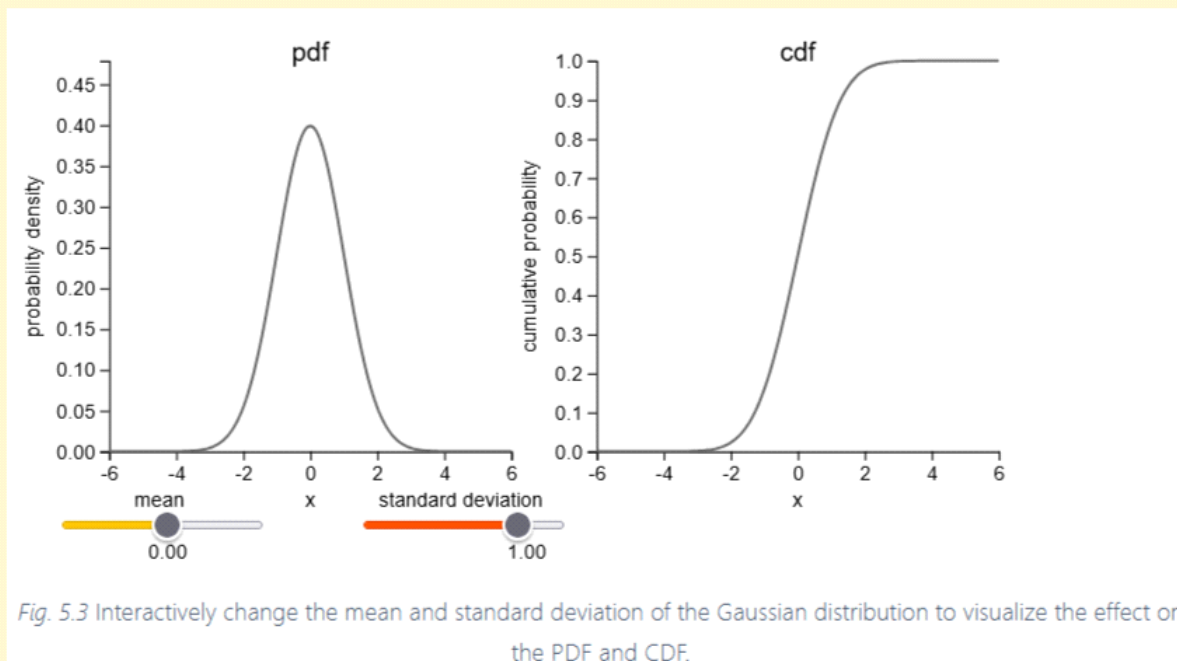


Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

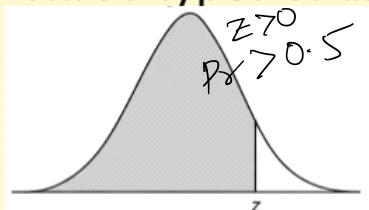
Gaussian



Normal distribution

- Three types of areas under the curve are useful

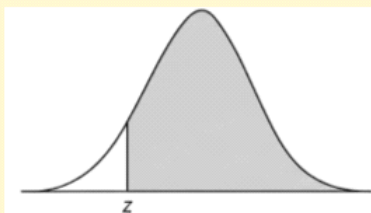
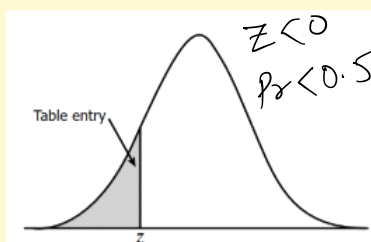
Confidence Interval



$$Pr(Z \leq z)$$

- Same as $\Phi(\cdot)$
- Read this from table
- check if $> \text{or} < 0.50$

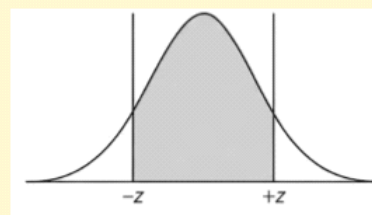
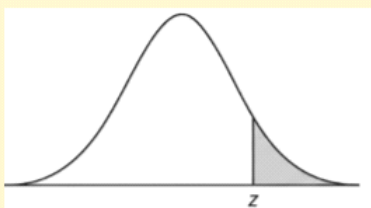
(CDF)



$$Pr(Z > z) = 1 - Pr(Z \leq z)$$

- Same as $1 - \Phi(\cdot)$
- check if $< \text{or} > 0.50$

(CCDF)

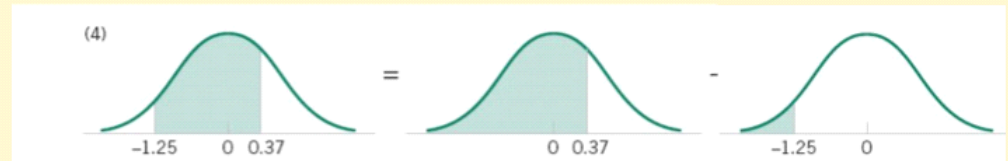
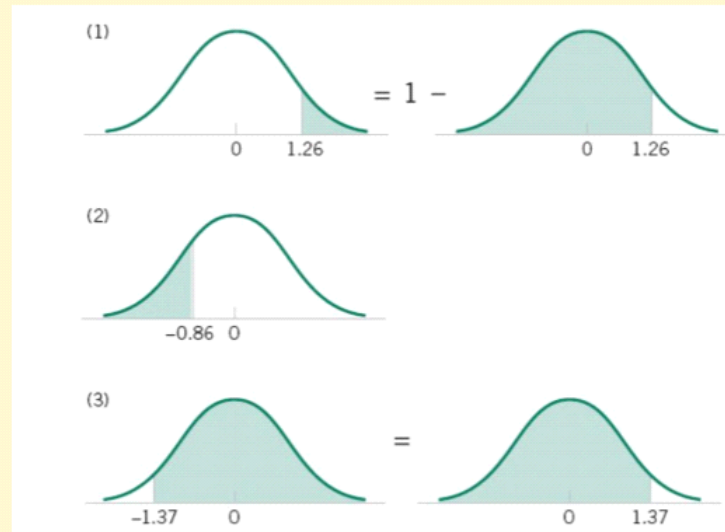


$$Pr(-z < Z < z)$$

$$\begin{aligned} &= Pr(Z \leq z) - Pr(Z \leq -z) \\ &= \Phi(z) - \Phi(-z) \\ &= 2\Phi(|z|) - 1 \end{aligned}$$

- 68-95-99.7 rule
 $\begin{array}{ccc} \longleftrightarrow & \longleftrightarrow & \longleftrightarrow \\ \pm 1\sigma & \pm 2\sigma & \pm 3\sigma \end{array}$

Examples (using basic prob. & symmetry)



Class problems

- **Example 3:** A concrete batch has a mean strength of 10 MPa; variance 4 MPa²
- What is the probability that a test result exceeds 13 MPa?

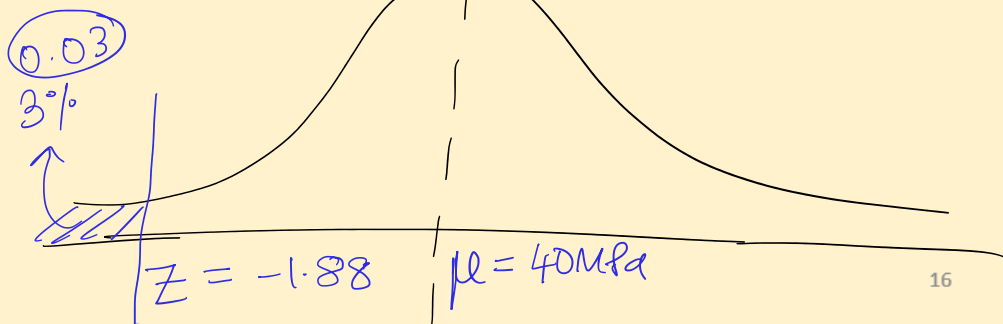
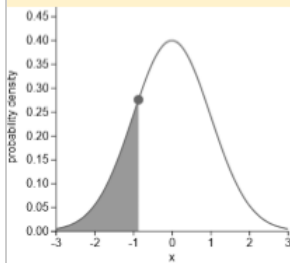
In-Class problem

- The compressive strength of a concrete mix is normally distributed with **mean 40 MPa** and **standard deviation 5 MPa**.

Out of 10,000 specimen produced, how many will generally have strength less than 35 MPa?

How many with strength less than 30 MPa?

Strength exceeded by 3,700 samples
 $\equiv 3.7\%$



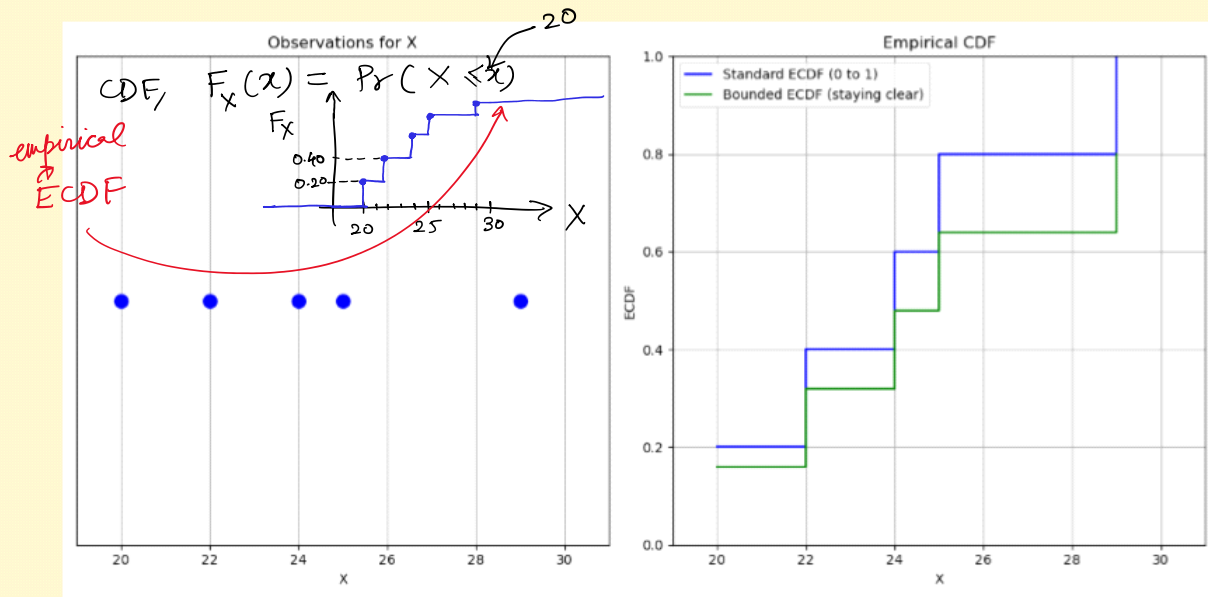
$$X = 40 + (-1.88) \times 5 = 30.60 \text{ MPa}$$

Inverse CDF

- For designing a structure, we often need a value that is not exceeded with more than p probability

$$x = F^{-1}(p)$$

Empirical distribution



Standard ECDF: $F_X(x) = \frac{i}{n}$, goes from 0 to 1.

Bounded ECDF: $F_{X_b}(x) = \frac{i}{n+1}$, stays clear of 0 & 1

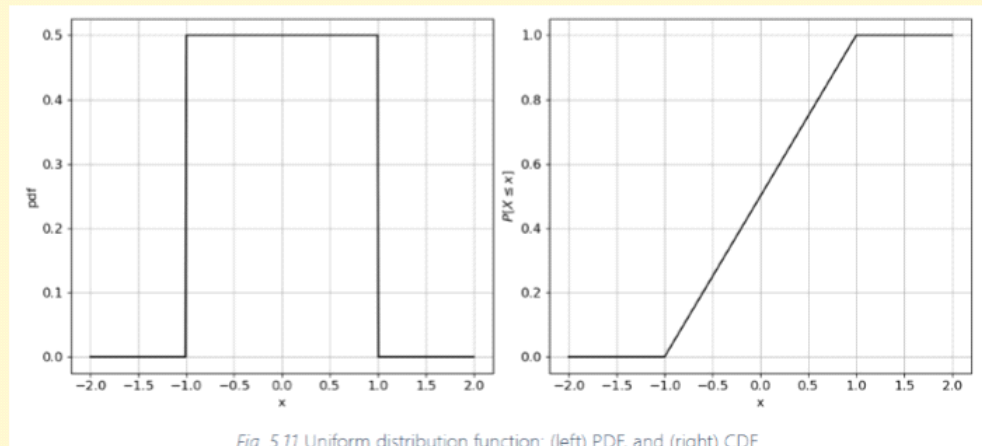
Lx
Sep yy, 2026

scipy

- Tutorial (not reference)
https://docs.scipy.org/doc/scipy/tutorial/stats/probability_distributions.html
- <https://docs.scipy.org/doc/scipy/tutorial/stats.html>
 - Run examples

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Uniform Distribution



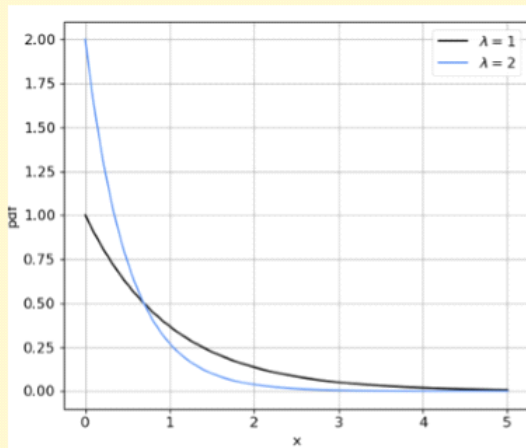
PDF

CDF

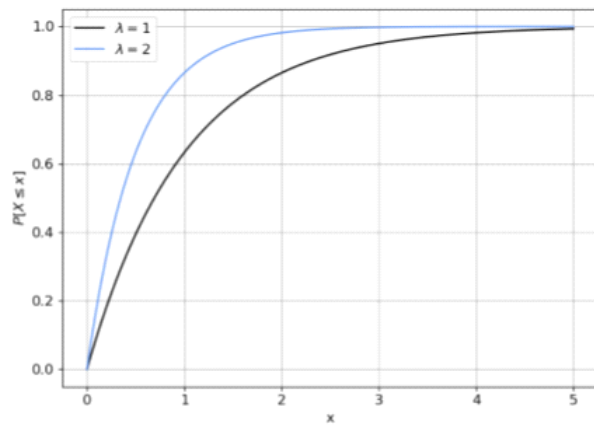
Properties

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Exponential Distribution



PDF



CDF

Properties

Lognormal Distribution: motivation

Lognormal Distribution: transformation

Lognormal Distribution: parameters

- A random variable is Lognormal, if its log follows Normal
 $X \sim \text{Lognormal} \Leftrightarrow Y = \ln(X) \sim \text{Normal}$

Knowing $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$, determine PDF of X

Gumbel distributions

- When we are interested in

the smallest

or

the largest of a set of rv's,

e.g., a chain of links: smallest strength.

Flood level under a bridge: highest flood level during its lifetime.



$$Y_1 =$$

$$Y_n =$$

Gumbel distributions

$$Y_1 = \min(X_1, X_2, \dots, X_n),$$
$$Y_n = \max(X_1, X_2, \dots, X_n).$$

The CDF of Y_1 is (smallest extreme):

$$F_{Y_1}(y) =$$

CDF

PDF

Gumbel distributions

$$Y_1 = \min(X_1, X_2, \dots, X_n),$$
$$Y_n = \max(X_1, X_2, \dots, X_n).$$

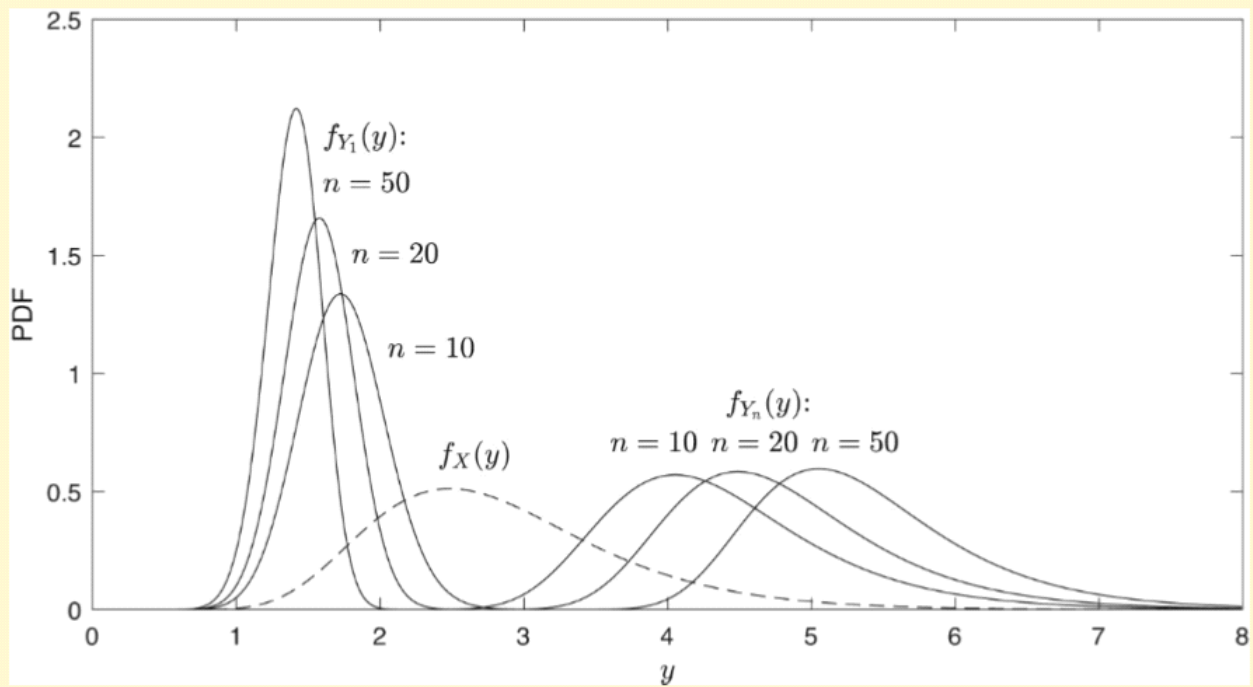
The CDF of Y_n is (largest extreme):

$$F_{Y_n}(y) =$$

CDF

PDF

Gumbel distributions



Gumbel distributions

Positive tail

PDF

CDF

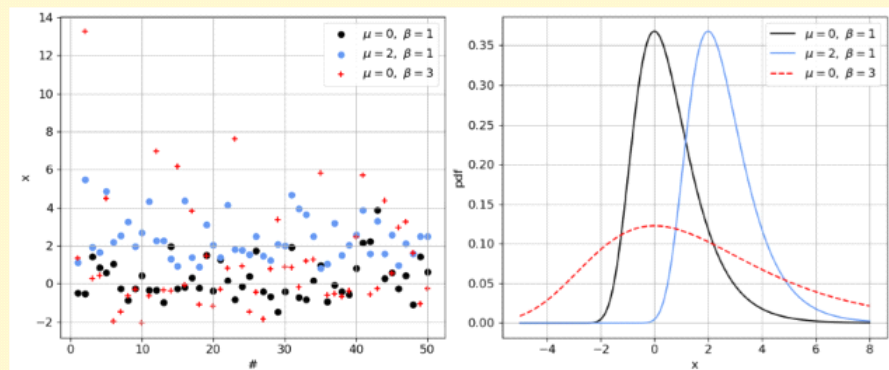


Fig. 5.14 Gumbel distribution function: (left) random samples, and (right) PDF. #

Negative tail

PDF

CDF

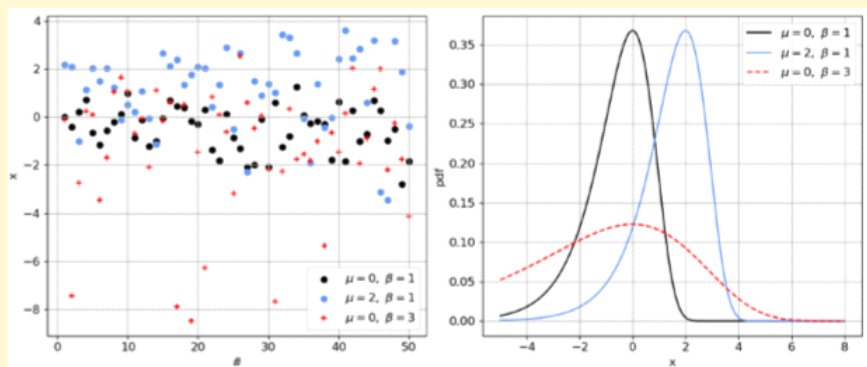
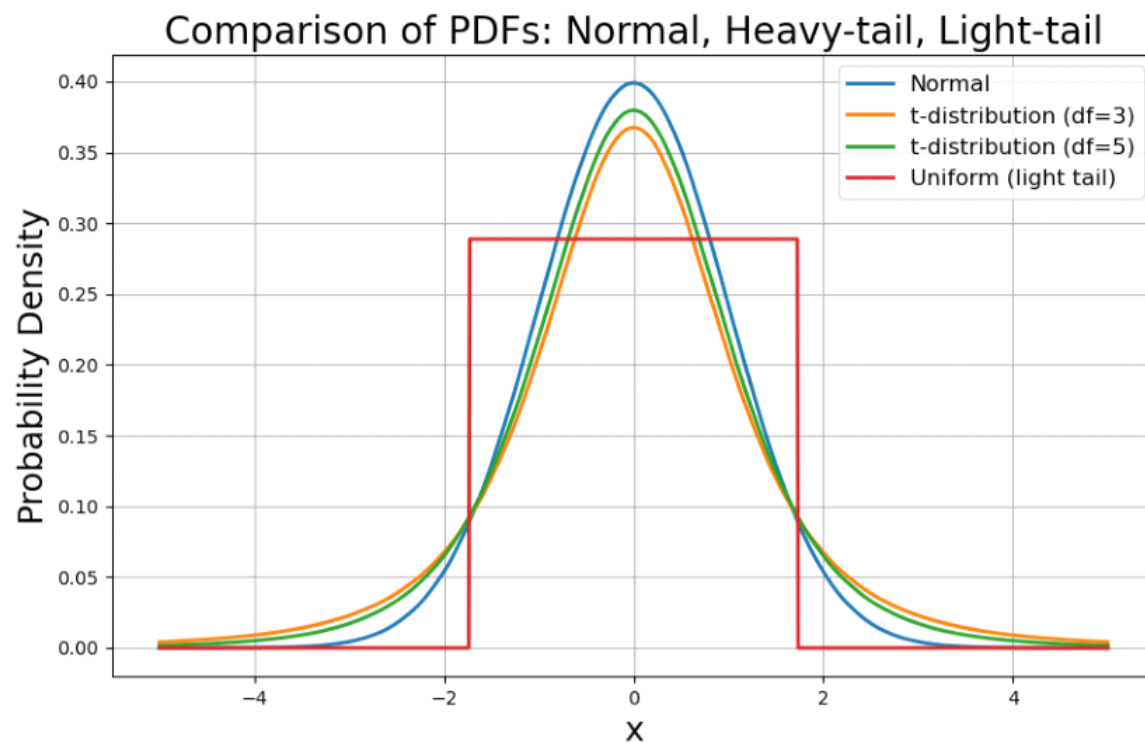


Fig. 5.16 Left-tailed Gumbel distribution function: (left) random samples, and (right) PDF. #

Core distributions

- *Tails*: thick/light



Next

B.1.3 Propagation laws &
least-squares
(Q1C3)

Questions, comments,
or concerns?